Independent branch polynomial review

Exact source: P = N^2 - b^2 F

F(a)=P(a)=0 implies N(a)=0

P'(a) = -b^2 F'(a) is nonzero

P and each factor are nonzero

Simple root: factor multiplicities sum to 1

Verdict: faithful_scoped (mathematical/source review).

The exact sign and scalar agree with source_norm_identity. Product nonzero is derived before multiplicity additivity. No division by2, b or the residual leading coefficient is used. Any field characteristic is allowed; only simple-root derivative nonvanishing is required.

The conclusion matches the three factor-nonzero conditions and multiplicity sum1 in source_support_base_multiplicity_sum. Source divisor congruences and determinant nonzero supply the identity. No separate geometric/DVR claim is inferred.

Rational fixture: coefficientData=(-4,0,0,-16,0), N=0 and U=-16X. Product=-16(X^5-X), derivative at0=16; multiplicities0,0,1. The residual leading coefficient is zero.

Controls, source bindings and review scope

Premise-removal controls (all over Q at0):

1. b=0: F=X, N=X, factors X,X,1. Product X^2 has order2.

2. Nonsimple F: F=X^2, b=1, N=0, factors X,X,-1. Product -X^2 has order2.

3. No support root: F=X, b=1, N=1, factors1-X,1,1. Product1-X has order0.

Each retains the exact generic identity. These tuples are not asserted to be realized by source coefficientData.

Characteristic2 witness: F=X, N=0, b=1, factors X,1,1. Since -X=X, the identity holds and product derivative is1.

Additional branch-root control: overQ, N=X+1, F=2X+1, b=1, productX^2. F(0)=1 and product order2.

Seven producer manifest entries and five source bindings match. Reviewer additionally binds coefficientData source and RingDivision.lean product-additivity API.

Reviewer authored ancestor replay plumbing/proof repair, not this direct derivation. Prospective review plan retained. No compiler, experiment or theorem replay by reviewer. The direct derivative route is source-level; current branch replay remains separate.

Machine-readable files: review.json, source-bindings.json and manifest.json. Detailed algebra: REPORT.md.